

```
plot2.m
1 % Define the grid for X1 and X2
2 [X1, X2] = meshgrid(-0.5:0.1:1.5, -0.5:0.1:1.5);
3
4 % Calculate the decision boundary plane for O1 (Z-axis)
5 % Equation:  $X1 + X2 - 2*O1 - 0.5 = 0 \Rightarrow 2*O1 = X1 + X2 - 0.5 \Rightarrow O1 = (X1 + X2 - 0.5) / 2$ 
6 O1_plane = (X1 + X2 - 0.5) / 2;
7
8 % Plot the plane
9 figure;
10 surf(X1, X2, O1_plane, 'FaceAlpha', 0.5, 'EdgeColor', 'none');
11 hold on;
12 xlabel('X1 Input');
13 ylabel('X2 Input');
14 zlabel('O1 Hidden Unit Output');
15 title('Decision Boundary Plane for O2:  $X1 + X2 - 2*O1 - 0.5 = 0$ ');
16 grid on;
17
18 % Plot the specific input points in (X1, X2, O1) space
19 % Points are formatted as [x1, x2, o1_val]
20 points = [0 0 0; 0 1 0; 1 0 0; 1 1 1];
21 plot3(points(:,1), points(:,2), points(:,3), 'ro', 'MarkerSize', 10, 'LineWidth', 2);
22 legend('Decision Plane', 'Input (00, 01, 10, 11)');
```